

Python Beginner Practice Problems

Randomized Practice Questions Covering Basics, OOP, and Standard Libraries

Mixed Practice Set (Beginner Level)

Problem 1: Square Root (Math Module)

Write a program that uses the `math` module to find and print the square root of a number entered by the user (e.g., 144).

Problem 2: Simple Class & Object (OOP)

Create a class named `Car` with a constructor that takes `brand` and `color`. Create an object from this class and print its attributes.

Problem 3: Current Year (Datetime Module)

Use the `datetime` module to get the current date and print only the current year.

Problem 4: Random Die Roll (Random Module)

Write a program using the `random` module to simulate rolling a standard 6-sided die (generating a random integer between 1 and 6).

Problem 5: Basic List & Loop (Python Basics)

Given a list of numbers: `[2, 4, 6, 8, 10]`, write a loop to print each number multiplied by 3.

Problem 6: Word Counter (Collections Module)

Given a list of colors: `['red', 'blue', 'red', 'green', 'blue', 'red']`, use `Counter` from the `collections` module to count how many times 'red' appears.

Problem 7: Simple Inheritance (OOP)

Create an animal base class named `Animal` with a method `speak()` that prints "Animal sound". Create a subclass `Dog` that inherits from it.

Problem 8: Formatting Dates (Datetime Module)

Write a program that prints today's date in the format: `Year/Month/Day` using the `datetime` module.

Problem 9: Even or Odd Function (Python Basics)

Write a function `check_even_odd(number)` that takes an integer and returns "Even" if it's divisible by 2, otherwise returns "Odd".

Problem 10: Random Choice (Random Module)

Given a list of fruit names: `['apple', 'banana', 'cherry', 'orange']`, use the `random` module to randomly select and print one fruit from the list.